

# GLOSSARY

## AAM (air-to-air missile)

A radar- or heat-guided missile fired at an enemy aircraft or missile, from another aircraft.

## Aerodynamics

The physics of the movement of objects through air or gas.

## Airfoil

See AIRFOIL *panel, right*.

## Aeronautics

The science of aerial travel.

## Aeronaut

The pilot of any lighter-than-air aircraft, especially a balloon.

## Afterburner

A device that injects additional fuel into a specially designed turbojet jetpipe to provide extra thrust. Also called reheat.

## Aileron

A control surface on the wing's far trailing edge – operated by moving the control column sideways – which governs roll.

## Airframe

An aircraft's body excluding engines.

## Altimeter

An instrument used to measure an aircraft's altitude.

## Angle of attack

The angle between the centerline of an aircraft's wing and the oncoming airflow. See also AIRFOIL *panel, right*.

## Antigravity suit

An antigravity or anti-G suit counteracts the effects of very fast flying by squeezing the legs and forcing blood to the brain so that a pilot does not lose consciousness.

## Artificial horizon

Primary cockpit flight instrument that indicates the aircraft's attitude in relation to the horizon. Also called a Gyro Horizon.

## Astronaut

A person engaged in space travel, derived from Greek words

meaning "star sailor." Cosmonaut is the Russian equivalent.

## Attitude

The tilt of an aircraft in relation to the oncoming airflow or to the ground.

## Autogyro

An aircraft equipped with a rotating wing, or rotor, to sustain itself in the air, and a propeller to move forwards. De la Cierva's machine was called an Autogyro.

## Automatic pilot

Airborne electronic system that automatically stabilizes an aircraft about its three axes, restores it to its original flight path after a disturbance, and can be preset to make the aircraft follow a particular flight path.

## AWACS

AWACS (Airborne Warning and Control System) refers to a modified Boeing airliner with a rotating radar dome (the E-3 Sentry), which provides all-weather

surveillance, command, and communications for allied air defense forces.

## Bank

To travel with one side higher than the other when turning. In an aircraft, the pilot must combine roll and yaw to achieve this manoeuvre. See also BASIC CONTROLS *panel, right*.

## Black box

A box – usually bright yellow or orange – containing electronic equipment, which records all the flight details. In the event of a crash, it is recovered from the aircraft's wreckage to ascertain the accident's cause.

## Cabane strut

Wing strut attached to the fuselage.

## Cantilever wing

A wing that does not require external struts or bracing wires.

## Chaff

Radar-reflective particulate matter

released by aircraft to confuse radar detection systems.

## Cockpit

The space in the fuselage occupied by a pilot (and/or crew). Usually refers to smaller aircraft.

## Cold War

Term given to the post-WWII period of tension between the two superpowers – the US and the USSR.

## Control column

Floor-mounted stick used by pilot to control roll (by moving it sideways) and pitch (by moving it forward or backward). Also called a control stick or joystick.

## Control surfaces

The moving parts of an aircraft that change the airflow around its trailing edges – i.e. ailerons, elevators, rudders – causing it to roll, pitch, or yaw.

## Cowling

Engine covering, often using hinged or removable panels.

## Cross-braced

Held rigid by wires attached to the extremities of the structural frame and crossing at its centre.

## Dihedral

The angling of the wing or tailplane upward from root to tip when seen from head on, to maintain lateral stability.

## Dirigible

An airship that is capable of being guided by an engine.

## Drag

The force (air resistance) that resists the motion of an aircraft.

## Drift indicator

An instrument indicating an aircraft's angle of "drift" – the difference between an aircraft's projected flight path and its actual heading, as affected by winds.

## Drone

A remotely piloted aircraft, used in a variety of military roles, including reconnaissance. Also known as an Unmanned Aerial Vehicle (UAV) or Remote Piloted Vehicle (RPV).

## Duralumin

Wrought alloy of aluminum with small percentages of copper, magnesium, and manganese.

## Ejection seat

A special seat on military aircraft which uses a rocket motor to blast the pilot clear in an emergency and parachute him or her to safety.

## ANATOMY OF AN AIRPLANE

